



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2008

MATHEMATICS EXTENSION 1

8:45am – 10:50am
Friday 5th September

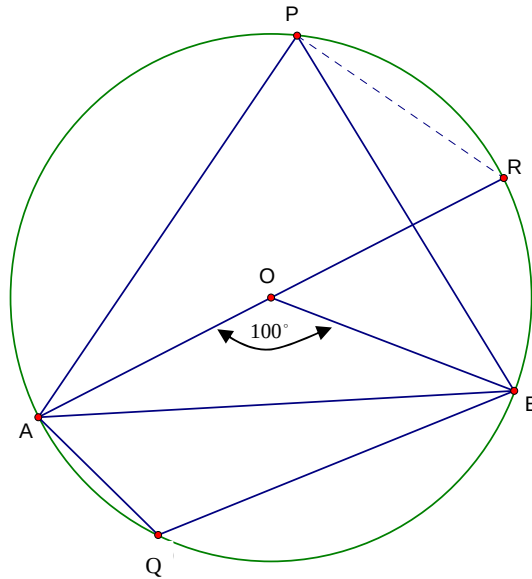
Directions to Students

• Reading Time: 5 minutes	• Total Marks: 84
• Working Time: 2 hours	
• Write using blue or black pen (sketches in pencil).	• Attempt Questions 1 – 7
• Board approved calculators may be used	• All questions are of equal value
• A table of standard integrals is provided at the back of this paper.	
• All necessary working should be shown in every question.	
• Answer each question in the booklets provided and clearly label your name and teacher's name.	

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QUESTION 1 (use a SEPARATE writing booklet)

- (a) If $x = \frac{nx_1 + mx_2}{m+n}$ and $x_1 = -2$ and $x_2 = 3$, find x when
- (i) $m = 3, n = 2$ (1M)
- (ii) $m = 3, n = -2$ (1M)
- (b) Noting that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, write down, without explanation, the value of:
- (i) $\lim_{x \rightarrow 0} \frac{\sin 2008x}{x}$ (1M)
- (ii) $\lim_{x \rightarrow 0} \frac{\sin(\pi - 2x)}{2008x}$, assuming that $\sin(\pi - 2x) = \sin 2x$. (1M)
- (c) Write down the derivative, with respect to x , of
- (i) $\cos^{-1} x$ (1M)
- (ii) e^{7x} (1M)
- (d) In the diagram, P, R, B, Q and A are points on the circumference of a circle centre O and AOR is a diameter.



- (i) Given $\angle AOB = 100^\circ$, briefly explain why $\angle APB = 50^\circ$ (1M)
- (ii) Write down, without explanation, the size of
- (a) $\angle AQB$ (1M)
- (b) $\angle BPR$ (1M)

- (e) Noting that $\frac{d}{dx}[e^x \sin x] = e^x \sin x + e^x \cos x$ evaluate

$$\int_0^{\frac{\pi}{2}} e^x (\sin x + \cos x) dx \quad (2M)$$

- (f) Given that if $y = 7^x$, then $\frac{dy}{dx} = A(7^x)$, where A is a constant,

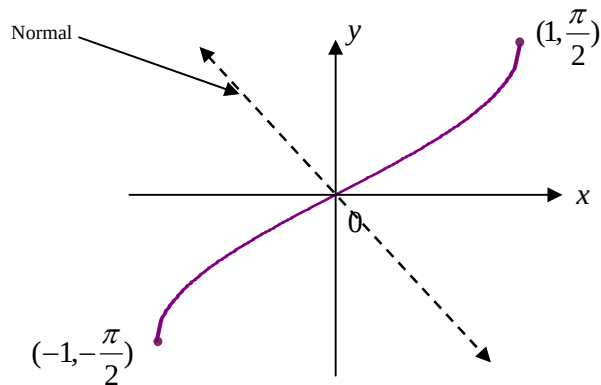
write down the exact value of A . (1M)

QUESTION 2 (use a SEPARATE writing booklet)

- (a) Write down the fifth term in the expansion by the binomial theorem of
(i) $(a + b)^8$ (1M)

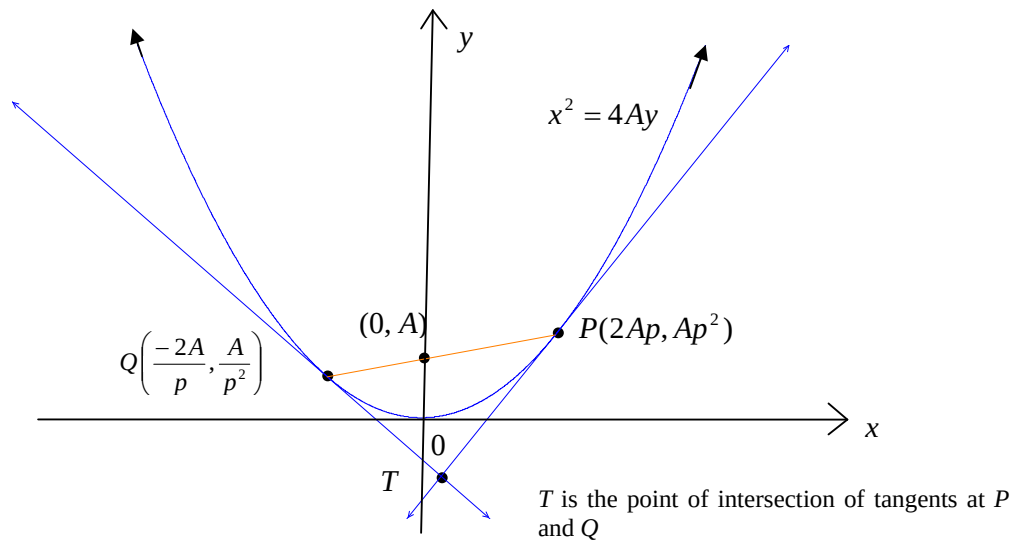
(ii) $(3x + \frac{1}{x})^8$ (2M)

- (b) Focus on the graph of $y = \sin^{-1} x$ below. State (answers only are required)



- (i) The domain of x . (1M)
- (ii) The gradient of the normal at O the origin. (1M)
- (iii) The equation of the horizontal line in which the curve $y = \sin^{-1} x$ may be reflected, to obtain a sketch of $y = \cos^{-1} x$. (1M)
- (c) Note that if $-\pi \leq \theta \leq \pi$, the solution to $\cos \theta = \frac{1}{2}$ is $\theta = \pm \frac{\pi}{3}$.
Assuming the above, write down in radians, the general solution of the equation $\cos \theta = \frac{1}{2}$. (2M)

(d)



PQ is a focal chord of the parabola

$$x^2 = 4Ay$$

The tangent at P has equation $y - px + Ap^2 = 0 \dots\dots\dots(\alpha)$.

The tangent at Q has equation $y + \frac{x}{p} + \frac{A}{p^2} = 0 \dots\dots\dots(\beta)$.

- (i) Do not prove the above but solve the equations (α) and (β) to show that the x- coordinate of T is $A(p - \frac{1}{p})$. (2M)
- (ii) Determine the y- coordinate of T. (2M)

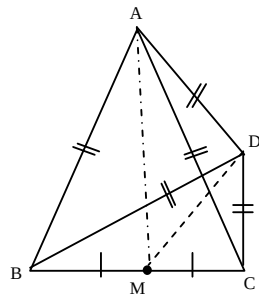
QUESTION 3 (use a SEPARATE writing booklet)

- (a) Consider the polynomial $P(x) = x^3 + Ax^2 - 2008$.

If $(x-2)$ is a factor of $P(x)$ find A . (2M)

- (b) Use the principle of mathematical induction to show that $(2008^n - 1)$ is divisible by 9 for all positive integers n . (4M)

- (c) The figure below ABCD is a regular tetrahedron, with

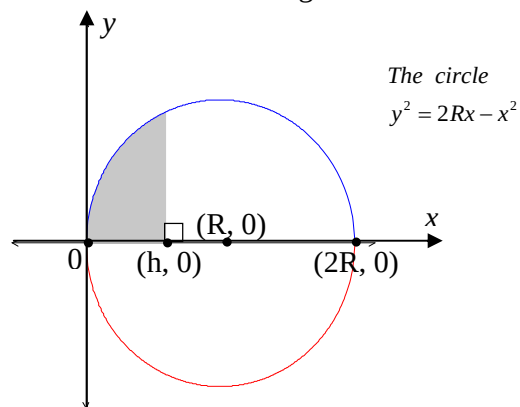


$AB = AC = BC = BD = DC = AD = 20\text{cm}$.

- (i) Draw the triangle BCD in your writing booklet and mark 'M' the midpoint of BC. Prove that $DM = 10\sqrt{3}\text{cm}$. (2M)

- (ii) Determine the size of $\hat{AMD}$. (2M)

- (d) A circle centre $(R, 0)$ and radius R has equation $y^2 = 2Rx - x^2$ and is drawn below. The shaded portion of the circle is rotated through 2π radians about the x -axis.



Prove that the volume V of the spherical cap generated is given by

$$V = \pi h^2 \left(R - \frac{h}{3}\right). \quad (2M)$$

QUESTION 4 (use a SEPARATE writing booklet)

- (a) A point P moves along the parabola whose equation is $y = \frac{x^2}{8}$.
The x -coordinate of P increases at the constant rate of 8cm/sec.
At what rate is the y -coordinate increasing when $x = 4$? (2M)

- (b) In solving a problem on Newton's Law of Cooling, Ignatius correctly arrives at the equation.

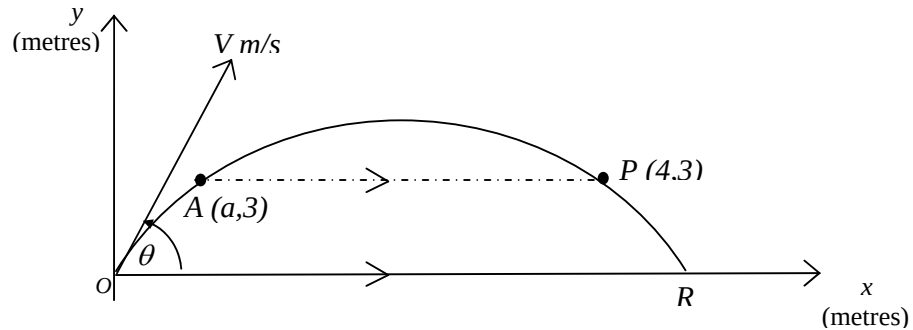
$$T = 20 + 980e^{-kt}$$

Where T is the temperature of a body at time t .

- (i) Given that at $t = 100$, $T = 510$, show $k = \frac{\ln 2}{100}$. (1M)
- (ii) Find T when $t = 200$. (1M)
- (c) If the equation $f(x) = 0$ has a root near $x = a$, it can be shown that, in general, a closer approximation to the root is a_1 , where $a_1 = a - \frac{f(a)}{f'(a)}$. Do not prove this result, known as Newton's method of approximating roots, but draw and label:
- (i) One diagram showing the regular situation in which a_1 is a better approximation than a . (2M)
- (ii) A second diagram, showing a case in which Newton's method does not apply, even though $f(x)$ has a root near $x = a$. (1M)

Question 4 continues on the next page.

- (d) A particle is projected from a point O with a speed of V m/s at an angle of θ to the horizontal. Air resistance is to be neglected and g ms⁻² is the acceleration due to gravity, vertically downwards.



In solving the following problems, you may assume that the Cartesian equation of the path of the projectile is given by:

$$y = \frac{-gx^2}{2V^2}(1 + \tan^2 \theta) + x \tan \theta. \text{ (Do NOT prove this result)}$$

You are given $V^2 = 8g$ and that the particle passes through a point $P(4,3)$.

- (i) Prove that $\tan \theta = 2$, without using the result of (ii) below. (2M)
- (ii) Assume that $\tan \theta = 2$, to show that the range $OR = 6.4$ m. (2M)
- (iii) The point $A = (a, 3)$ lies on the trajectory, where AP is parallel to the x-axis. What is the value of a ? (1M)

QUESTION 5 (use a SEPARATE writing booklet)

- (a) Noting that $\log_e e^N = N$ show by setting $u = \log_e x$, that (3M)

$$2 \int_{e^2}^{e^5} \frac{\log_e x}{x} dx = 21$$

- (b) Find the values of x and y which satisfy the simultaneous equations

$$\cos^{-1} x + 3 \sin^{-1} y = \pi \dots\dots(\alpha) \quad (3M)$$

$$\cos^{-1} x - 3 \sin^{-1} y = 0 \dots\dots(\beta)$$

- (c) Here you may use, without proof, the identity

$$\sin 3A = 3 \sin A - 4 \sin^3 A \dots\dots\dots(\alpha)$$

- (i) By direct substitution show α is true if $A = \frac{\pi}{6}$. (1M)

- (ii) Use α to find, in exact form, the smallest positive value of x

$$\text{which satisfies the equation } 6x - 8x^3 = 1. \quad (3M)$$

(suggestion: let $x = \sin A$)

- (d) Assuming that

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \dots\dots\dots\delta$$

factorise

$$(\alpha - \beta)^3 + (\beta - \gamma)^3 + (\gamma - \alpha)^3. \quad (2M)$$

QUESTION 6 (use a SEPARATE writing booklet)

- (a) A particle, executing Simple Harmonic Motion, is moving along the x -axis with velocity v , and acceleration $\ddot{x}$.

(i) Show that $\ddot{x} = \frac{d}{dx} \left(\frac{v^2}{2} \right)$. (1M)

- (ii) Assuming that in the usual notation $\ddot{x} = -n^2 x$ and that the particle is released from rest from the point where $x = a$, and at zero time, show that:

(α) $v = -n\sqrt{a^2 - x^2}$. (2M)

(β) $x = a \cos nt$. (2M)

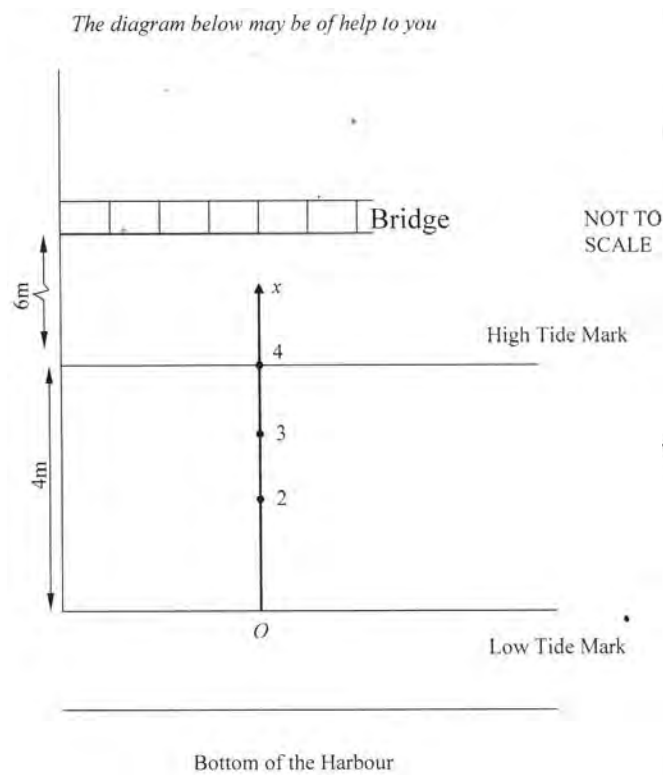
- (b) In Aloha Harbour the height of the water above the low tide mark is given by the rule $x = 2(1 + \cos \theta)$ metres, where $\theta = \frac{\pi t}{390}$ radians and t is the time in minutes after high tide. (see the diagram on the next page)

Explain why the time interval between high and low tide is 390 minutes. (2M)

- (c) Given $t = 65$, show that the rate at which the water level is falling is approximately 8mm per minute. (2M)

Question 6 continues on the next page.

(d)



The Aloha Harbour Bridge is 10 metres above the low water mark.

A pleasure yacht can just sail under the bridge when the distance between the bridge and the water is 7 metres. How long after high tide will it be before the yacht can first sail under the bridge?

(3M)

QUESTION 7 (use a SEPARATE writing booklet)

- (a) You may assume that $x^4(1-x)^4 = x^8 - 4x^7 + 6x^6 - 4x^5 + x^4$

Let $J = \int_0^1 x^4(1-x)^4 dx$.

Show that $J = \frac{1}{630}$ (3M)

- (b) Let $E = \frac{x^4(1-x)^4}{x^2+1}$.

Noting the assumption in part (a) and using the process of long division for polynomials, show that

$$E = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2} \quad (3M)$$

- (c) Let $I = \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$. Use the result in part (b) to prove that (3M)

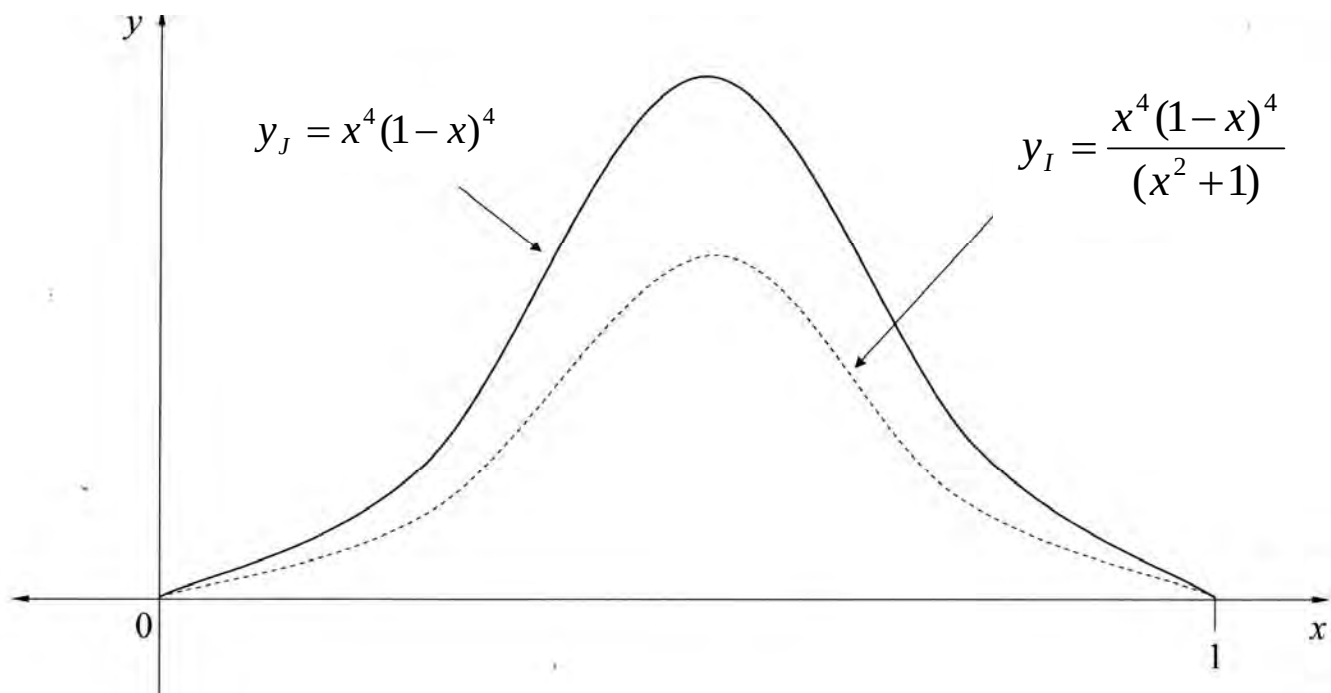
$$I = \frac{22}{7} - \pi.$$

- (d) Assuming, without any explanation, that $J > I > \frac{J}{2}$, deduce from (a) and (c) that

$$\frac{22}{7} - \frac{1}{630} < \pi < \frac{22}{7} - \frac{1}{1260}. \quad (2M)$$

Question 7 continues on the next page.

(e)



Use the diagram above, or otherwise, to explain why $J > I$.

(1M)



Suggested
Solutions

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Solutions, Marking Scheme & Comments

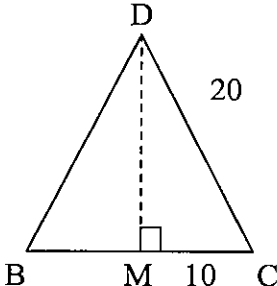
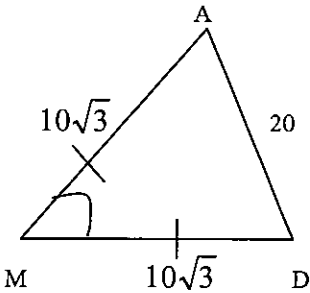
QUESTION 1			
Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) (i) $x_{(i)} = \frac{2 \times (-2) + (3 \times 3)}{(3 + 2)} = \frac{5}{5} = 1$ (ii) $x_{(ii)} = \frac{(-2 \times -2) + (3 \times 3)}{(3 - 2)} = \frac{13}{1} = 13$ <i>(Award the mark if the candidate makes just one error)</i>	1 1		
(b) (i) $E_{(i)} = (2008) \lim_{x \rightarrow 0} \left(\frac{\sin 2008x}{2008x} \right) = 2008 \times 1 = 2008$ (ii) $E_{(ii)} = \lim_{x \rightarrow 0} \left(\frac{\sin 2x}{2x} \right) \times \frac{1}{1004} = \frac{1}{1004}$	1 1		
(c) (i) $\frac{-1}{\sqrt{1-x^2}}, x \leq 1$ (ii) $7e^{7x}$	1 1		
(d) (i) In the given circle, $\hat{AOB}$ is the angle at the centre, subtended by the arc AQB. $\hat{APB}$ is the angle at the circumference of the same circle, subtended by the arc AQB. $\therefore \hat{APB} = \frac{1}{2} \hat{AOB} = \frac{100^\circ}{2} = 50^\circ$. (ii) (a) 130° (b) 40°	1 1 1		
(e) $I = \left[e^x \sin x + e^x \cos x \right]_0^{\frac{\pi}{2}}$ $= \left[e^{\frac{\pi}{2}} + 0 \right] - [0 + 1]$ $= e^{\frac{\pi}{2}} - 1$ <i>(No penalty for one error)</i>	2		
(f) $A = \ln 7$	1		

QUESTION 2			
Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) (i) $T_5 = \binom{8}{4} a^4 b^4$ (ii) $T_5 = \binom{8}{4} (3x)^4 \left(\frac{1}{x}\right)^4 = \frac{8.7.6.5}{4.3.2.1} 81.x^0 = 5670$ (in part (ii) no penalty for one error)	1 2		
(b) (i) $-1 \leq x \leq 1$ (ii) -1 (iii) $y = \frac{\pi}{4}$	1 1 1		
(c) General Solution is $\theta = 2\pi n \pm \frac{\pi}{3}$ where n is an integer, positive, negative or zero. Award 1 mark for giving some new correct value of θ in radians.	2		
(d) (i) Eliminate y to get $x\left(p + \frac{1}{p}\right) - A\left(p^2 - \frac{1}{p^2}\right) = 0$ So $x\left(p + \frac{1}{p}\right) = A\left(p^2 - \frac{1}{p^2}\right)$ $= A\left(p - \frac{1}{p}\right)\left(p + \frac{1}{p}\right)$$\therefore x = A\left(p - \frac{1}{p}\right)$ (ii) Substitute $x = A\left(p - \frac{1}{p}\right)$ into $y = px - Ap^2$ to get $y = p\left[Ap - \frac{A}{p}\right] - Ap^2 = Ap^2 - A - Ap^2 = -A$ The y coordinate of T is $-A$	2 2		

QUESTION 3

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) Let $f(x) = x^3 + Ax^2 - 2008$ then $f(2) = 0$ $\therefore 0 = 8 + 4A - 2008 \therefore A = 500$	2		
(b) Step 1: When $n = 1$, $2008^1 - 1 = 2007 = 223 \times 9$, Which is divisible by 9, and Thus the statement is true for $n = 1$. Step 2: Assume the statement true for $n = k$ i.e. assume that $2008^k - 1$ is divisible by 9, i.e. that $\frac{2008^k - 1}{9}$ is some integer M. i.e. $\frac{2008^k - 1}{9} = M$, i.e. $2008^k - 1 = 9M \Rightarrow 2008^k = 9M + 1 \dots \dots **$ We now prove that the statement is true for $n = k + 1$ i.e. $2008^{k+1} - 1$ is divisible by 9 Now $2008^{k+1} - 1 = 2008^1 \cdot 2008^k - 1$ $= 2008(9M + 1) - 1 \dots \dots **(\text{by our assumption})$ $= 9 \times 2008M + 2008 - 1$ $= 9 \times 2008M + 2007$ $= 9(2008M + 223)$ $= 9I$ (where I is a positive integer) which is divisible by 9 Thus, if $2008^k - 1$ is divisible by 9 when $n = k$, then it is also divisible by 9 when $n = k + 1$ Step 3: Since the statement is true for $n = 1$, it is true for $n = 1 + 1 = 2$, and $n = 2 + 1 = 3$, and so on for all positive integral values of n.	4		

QUESTION 3

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(c) (i)  From symmetry, $\hat{DMC} = 90^\circ$ In right angle triangle DMC $DM^2 = 20^2 - 10^2 = 300$ $= 100 \times 3$ $\therefore DM = 10\sqrt{3} \text{ cm}$	2		
(c) (ii)  Apply cosine rule to $\triangle AMD$. $\cos \hat{AMD} = \frac{300 + 300 - 400}{600} = \frac{1}{3}$ $\therefore \hat{AMD} = \cos^{-1}\left(\frac{1}{3}\right) \approx 70^\circ 32'$	2		
(d) $V = \pi \int_0^h (2Rx - x^2) dx$ $= \pi \left[Rx^2 - \frac{x^3}{3} \right]_0^h$ $= \pi \left[Rh^2 - \frac{h^3}{3} \right] - [0]$ $= \pi h^2 \left[R - \frac{h}{3} \right]$	2		

(a)

When $x = 4$, $\frac{dy}{dt} = \left(\frac{4}{4} \times 8 \right) = 8 \text{ cm/sec}$

2

(i) Substitute values given to get $510 = 20 + 980e^{-100K}$

$$\therefore e^{-100K} = \frac{490}{980} = \frac{1}{2} \longrightarrow \therefore K = \frac{\ln 2}{100}$$

(ii) Here $T = 20 + 980e^{-2 \ln 2}$

$$= 20 + 245$$

$$= 265$$

1

1

(i) See Page 14.

(ii) See Pages 15 and 16.

2

1

(i) Into the given equation, substitute $V^2 = 8g$, $x = 4$, $y = 3$ (data) to get

$$3 = \frac{-16g}{2 \times 8g} (1 + \tan^2 \theta) + 4 \tan \theta$$

$$3 = -1 - \tan^2 \theta + 4 \tan \theta$$

So $\tan^2 \theta - 4 \tan \theta + 4 = 0$

$$\therefore (\tan \theta - 2)^2 = 0 \text{ i.e. } \tan \theta = 2 \text{ (twice)}$$

Note. That no other value of $\tan \theta$ satisfies.

2

(ii) Set $y = 0$, to get

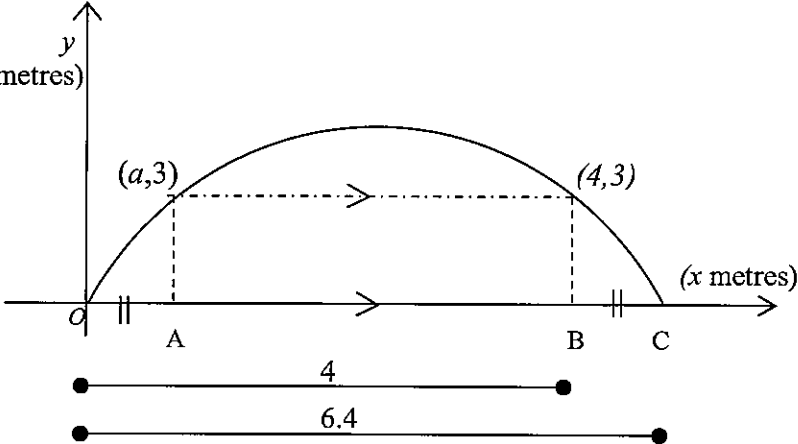
$$0 = \frac{-x^2}{16}(1+4) + 2x$$

$$\therefore \frac{x}{16}(5x-32)=0$$

$$\therefore x = 0 \text{ or } x=6.4$$

2

QUESTION 4

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(d) (iii)  From symmetry, $OA = a = BC$ Now $OC = 6.4$ $BC = OC - OB$ $= 6.4 - 4$ $= 2.4$ $\therefore OA = a = BC$ $\therefore a = 2.4$	1		

QUESTION 5

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) let $u = \log_e x$, then $du = \frac{dx}{x}$ Now if $x = e^5$, $\log_e x = 5$ $x = e^2$, $\log_e x = 2$ Now $I = 2 \int_{e^2}^{e^5} \log_e x \frac{dx}{x}$ $= 2 \int_2^5 u du = \left[u^2 \right]_2^5 = 25 - 4 = 21$	3		
(b) Add α and β to get $2 \cos^{-1} x = \pi$ $\therefore \cos^{-1} x = \frac{\pi}{2}$, giving $x = 0$. Using β, $\frac{\pi}{2} - 3 \sin^{-1} y = 0$, $-3 \sin^{-1} y = -\frac{\pi}{2}$ $\sin^{-1} y = \frac{\pi}{6} \therefore y = \frac{1}{2}$	3		
(c) (i) L.H.S = $\sin 3A = \sin \frac{\pi}{2} = 1$, if $A = \frac{\pi}{6}$ R.H.S = $3 \sin \frac{\pi}{6} - 4 \sin^3 \frac{\pi}{6} = \frac{3}{2} - \frac{1}{2} = 1$ So L.H.S = R.H.S = 1 if $A = \frac{\pi}{6}$ (ii) Given equation can be written as $3x - 4x^3 = \frac{1}{2}$ or $3 \sin A - 4 \sin^3 A = \frac{1}{2}$, where $x = \sin A$ $\therefore \sin 3A = \frac{1}{2}$ giving smallest positive value of $3A = \frac{\pi}{6}$ $\therefore A = \frac{\pi}{18}$ is the required answer.	1 3		

QUESTION 5

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(d) Given, $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) \dots (\delta)$ Let $a = (\alpha - \beta), b = (\beta - \gamma), c = (\gamma - \alpha)$ in the given identity δ. Then $a + b + c = (\alpha - \beta) + (\beta - \gamma) + (\gamma - \alpha) = 0$, so R.H.S of $\delta = 0$. $\therefore a^3 + b^3 + c^3 - 3abc = 0$ $\therefore a^3 + b^3 + c^3 = 3abc$ ie $(\alpha - \beta)^3 + (\beta - \gamma)^3 + (\gamma - \alpha)^3 = 3(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$ $\therefore$ The required factors are $3(\alpha - \beta)(\beta - \gamma)(\gamma - \alpha)$	2		

QUESTION 6

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) (i) Many approaches are possible for example $ \begin{aligned} RHS &= \frac{1}{2} \frac{d}{dx}(v^2) = \frac{1}{2} 2v \frac{dv}{dx} \\ &= \frac{dx}{dt} \cdot \frac{dv}{dx} = \frac{dv}{dt} = \text{acceleration} \end{aligned} $	1		
(ii) (α) From part (i) $\frac{d}{dx}\left(\frac{v^2}{2}\right) = -n^2 x$, (where $n = \frac{2\pi}{T}$, T being the period). Integrate both sides with respect to x to get, $\frac{v^2}{2} = -\frac{n^2 x^2}{2} + B. \quad (B \text{ the constant of integration})$ From data, when $t = 0$, $v = 0$ and $x = a$ $\therefore 0 = -\frac{n^2 a^2}{2} + B \Rightarrow B = \frac{n^2 a^2}{2} \Rightarrow \frac{v^2}{2} = -\frac{n^2 x^2}{2} + \frac{n^2 a^2}{2}.$ Multiply by 2 to arrive at , $v^2 = n^2 (a^2 - x^2), \quad \text{so } v = \pm n\sqrt{a^2 - x^2}.$ But when $t = 0$, $x = a$ (data), so particle must begin by moving to the left. (Note $x = a$ is the maximum positive position of P). $\therefore v = -n\sqrt{a^2 - x^2}, \quad x \leq a$	2		
(ii) (β) We can re-arrange the result of part (ii) (α) viz, $\frac{dx}{dt} = -n\sqrt{a^2 - x^2}$, $x \leq a$, to get $-\int \frac{dx}{\sqrt{a^2 - x^2}} = \int n dt$ $\therefore \cos^{-1}\left(\frac{x}{a}\right) = nt + C. \quad (C \text{ is the constant of integration}).$ But from data, when $t = 0$, $x = a$. $\therefore C = 0$, as $\cos^{-1}(1) = 0$ $\therefore \cos^{-1}\left(\frac{x}{a}\right) = nt \Rightarrow \text{So } x = a \cos nt$	2		

QUESTION 6

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(b) Here $x = 2 + 2 \cos\left(\frac{\pi}{390}\right)t$, the standard S.H.M. form $x = a \cos nt. \text{ So } n = \left(\frac{\pi}{390}\right) \text{ and period} = \frac{2\pi}{n} = 780 \text{ mins}$ So from high tide to next high tide = 780 mins. Hence from high tide to next low tide = 390 mins.	2		
(c) When $t = 65$, $\frac{dx}{dt} = -\frac{\pi}{195} \sin \frac{\pi t}{390} = -\frac{\pi}{195} \sin \frac{\pi 65}{390} = \left(-\frac{\pi}{390} 1000\right) \text{ mm/min}$ $\approx -8.055...(\text{calc}) \approx -8 \text{ mm/min.}$ The minus sign indicates that the water level is falling.	2		
(d) Since the distance between the bridge and the water has to be 7m, clearly $x=3$ $\therefore 2(1 + \cos \frac{\pi t}{390}) = 3, \text{ giving } \cos \frac{\pi t}{390} = \frac{1}{2}$ $\therefore \frac{\pi t}{390} = \frac{\pi}{3}. \text{ Hence } t = 130$ The yacht can travel under the bridge 2 hours 10 mins after high tide.	3		

QUESTION 7

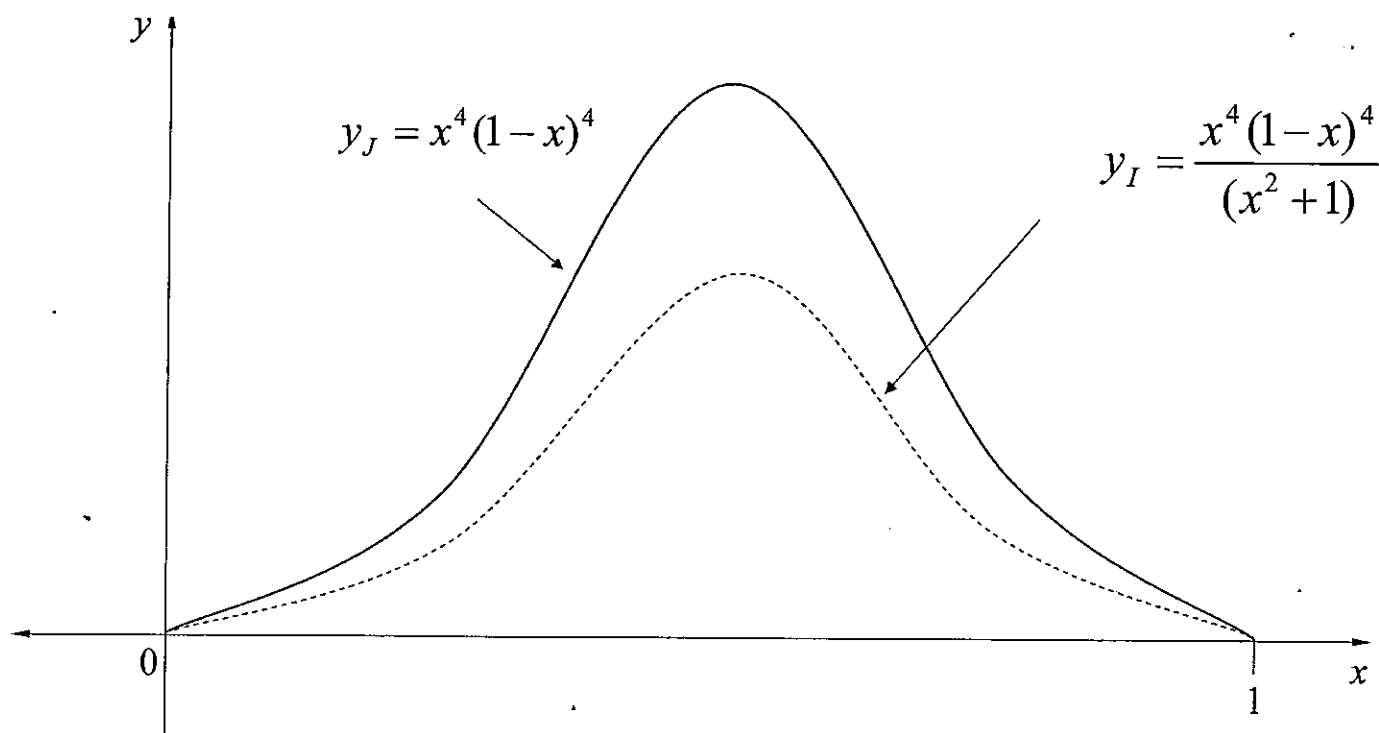
Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) $J = \int_0^1 x^4(1-x)^4 dx$ $= \int_0^1 (x^8 - 4x^7 + 6x^6 - 4x^5 + x^4) dx \dots\dots\dots (given)$ $= \left[\frac{x^9}{9} - \frac{x^8}{2} + \frac{6x^7}{7} - \frac{4x^6}{6} + \frac{x^5}{5} \right]_0^1$ $= \frac{1}{9} - \frac{1}{2} + \frac{6}{7} - \frac{2}{3} + \frac{1}{5}$ $= \frac{70 - 315 + 540 - 420 + 126}{630}$ $= \frac{736 - 735}{630} = \frac{1}{630}$	3		
(b) First do the long division $\begin{array}{r} x^6 - 4x^5 + 5x^4 - 4x^2 + 4 \\ x^2 + 1 \overline{) x^8 - 4x^7 + 6x^6 - 4x^5 + x^4} \\ \underline{x^8 + x^6} \\ -4x^7 + 5x^6 - 4x^5 + x^4 \\ \underline{-4x^7 - 4x^5} \\ +5x^6 + x^4 \\ \underline{+5x^6 + 5x^4} \\ -4x^4 \\ \underline{-4x^4 - 4x^2} \\ 4x^2 \\ \underline{4x^2 + 4} \\ -4 \end{array}$ $\therefore E = x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{x^2 + 1}$	3		

QUESTION 7

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(c) $I = \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx$ $= \int_0^1 (x^6 - 4x^5 + 5x^4 - 4x^2 + 4 - \frac{4}{1+x^2}) dx \dots (\text{from part b})$ $= \left[\frac{x^7}{7} - \frac{4x^6}{6} + x^5 - \frac{4x^3}{3} + 4x - 4 \tan^{-1} x \right]_0^1$ $= \left[\frac{1}{7} - \frac{2}{3} + \frac{5}{5} - \frac{4}{3} + 4 \right] - 4[\tan^{-1}(1)]$ $= \frac{30 - 140 + 210 - 280 + 840}{210} - 4\left(\frac{\pi}{4}\right)$ $= \frac{1080 - 420}{210} - \pi$ $= \frac{660}{210} - \pi$ $= \frac{22}{7} - \pi$	3		
(d) Given that $J > I > \frac{J}{2}$ $\frac{1}{630} > \frac{22}{7} - \pi > \frac{1}{1260} \text{ from parts (a) and (c)}$ (multiply through by -1) $-\frac{1}{630} < \pi - \frac{22}{7} < -\frac{1}{1260}$ (add $\frac{22}{7}$ to each expression) $\frac{22}{7} - \frac{1}{630} < \pi < \frac{22}{7} - \frac{1}{1260}$	2		

QUESTION 7

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(e) From observation J represents the area under the full curve. I represents the area under the broken curve. So clearly $J > I$.	1		



An Algebraic Proof:

For $0 < x < 1$ we have

$$1 < 1 + x^2 < 2$$

$$\text{So } x^4(1-x)^4 > \frac{x^4(1-x)^4}{1+x^2} > \frac{x^4(1-x)^4}{2}$$

$$J > I > \frac{J}{2}$$

Source of Question 7:

N.S.W. Leaving Certificate Maths 1 Honours Paper 1964, Q9.

Also adapted from a Cambridge Matriculation examination paper, late 1930's.

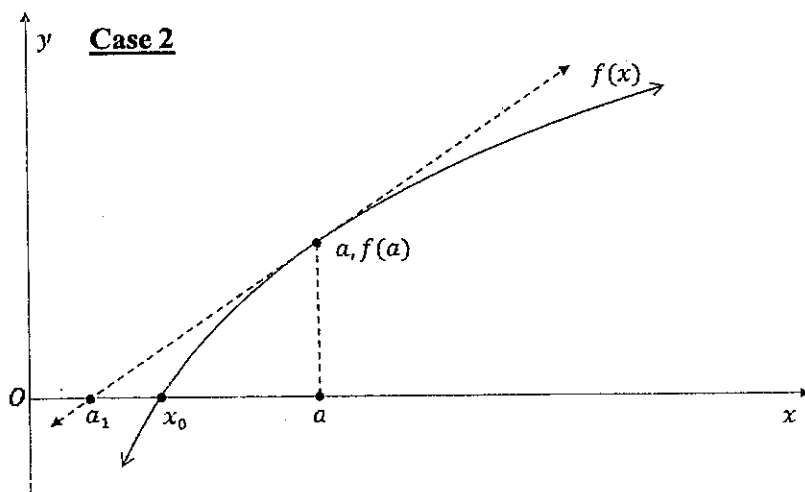
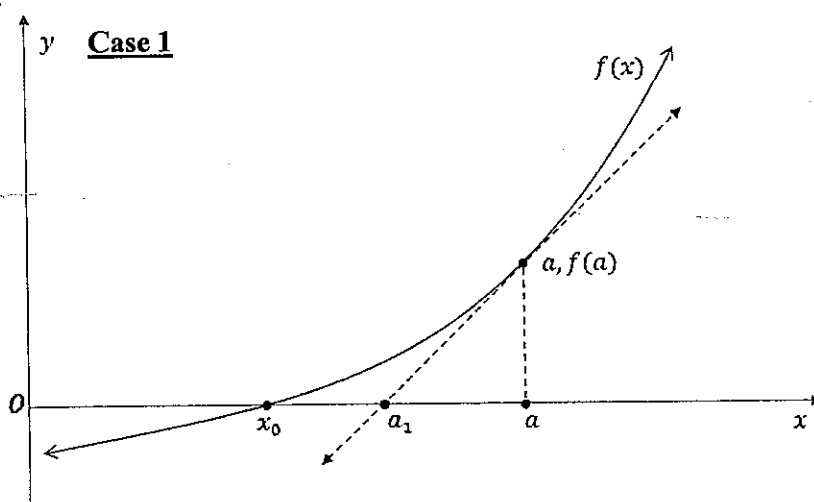
Question (4C)

In the following examples the required root of $f(x)=0$ is given by x_0 .

In cases 1 and 2 below, Newton's method works and a_1 is a better approximation than a .

(i)

(2 marks)

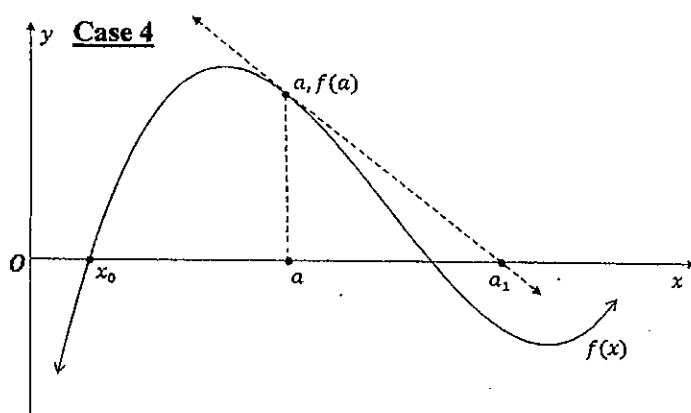
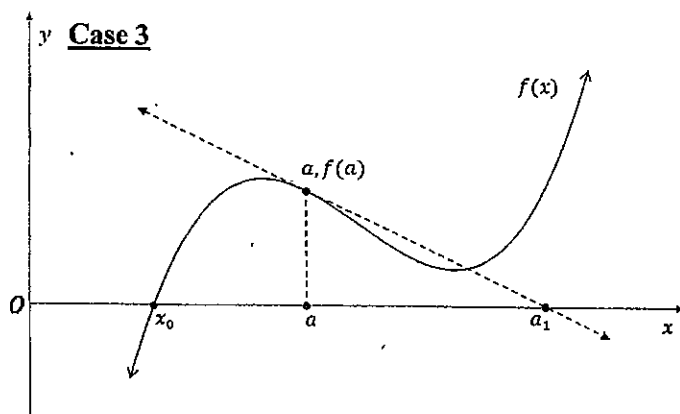


Question (4C)

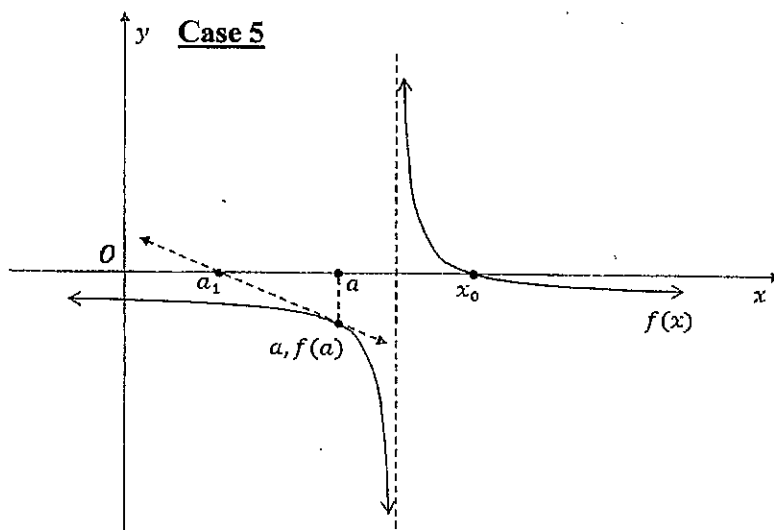
In cases 3, 4 and 5, Newton's method fails and a_1 is not a better approximation than a .
For cases 3 and 4, this is because $f'(x) = 0$ at some point between a and x_0 or at $x = a$

(ii)

(1 mark)

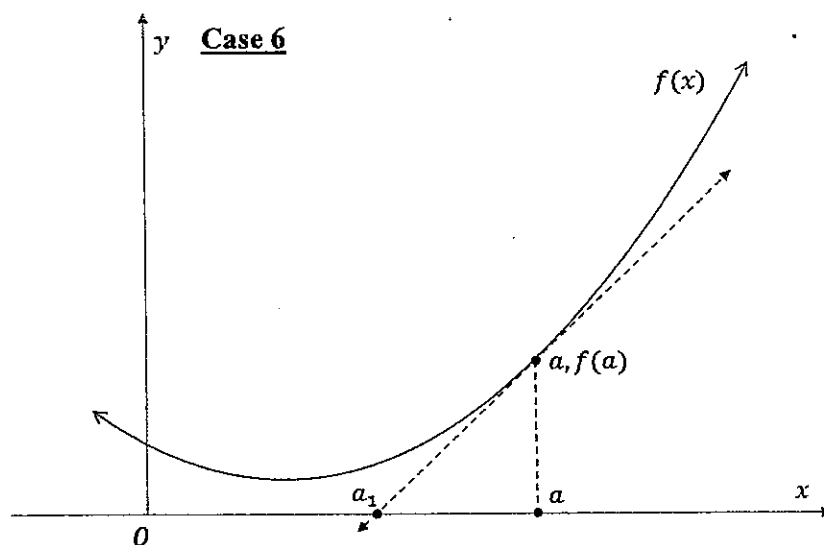


For case 5, $f'(x)$ is undefined at some point between a and x_0



Question (4C)

The original question asks to show a case where Newton's method does not apply "even though $f(x)$ has a root near $x = a$ ". Case 6 shows a function $f(x)$ that does not have a root at all. Hence, this case is invalid as a solution.



Case 7 below shows a function similar to that in case 1. However, this case illustrates that the first approximation using Newton's method, a_1 is not necessarily closer than the initial approximation a . However, subsequent approximations ($a_2, a_3, \dots$) will get progressively closer to the required root x_0 . This is not necessarily the case in cases 3, 4 and 5 where Newton's method fails.

